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TECHNICAL PAPER

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Observations and Interpretations from the Canadian Fine Particle Monitoring Program

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ABSTRACT

Canadian particle monitoring programs examining PM_{10}, $PM_{2.5}$, and particle composition have been in operation for over 10 years. Until recently, the measurements were manual/filter-based with 24-hr sample collection varying in frequency from daily to every sixth day, using Graseby-Anderson dichotomous samplers. In the past few years, these monitoring activities have been expanded to include hourly measurements using tapered element oscillating microbalances (TEOMs). This continuous monitoring program started operation focusing on PM_{10}, but now emphasizes $PM_{2.5}$ through the addition of more TEOMs and switching of the inlets of some of the existing units. The data from all of these measurement activities show that there are broad geographical differences and also local- to regional-scale spatial differences in mass and composition of $PM_{2.5}$. Due to variations in sources, significantly different $PM_{2.5}$ concentrations are not uncommon within the same city. Comparison of nearby urban and rural sites indicates that 30 and 40% of the $PM_{2.5}$ is from local urban sources in Montreal and Toronto, respectively. Hourly $PM_{2.5}$ measurements in Toronto suggest that vehicular emissions are an important contributor to urban $PM_{2.5}$. There has been a decreasing trend in urban $PM_{2.5}$, with annual average concentrations between the 1987–1990 and 1993–1995 periods decreasing by 11 to 39%, depending upon the site. The largest declines were in Montreal and Halifax, and the smallest decline was in Toronto. Comparison of 24-hr TEOM and manual dichotomous sampler $PM_{2.5}$ measurements from a site in Toronto indicates that the TEOM results in lower concentrations. The magnitude of this difference is relatively small in the warmer months, averaging about 12%. During the colder months the difference averages about 23%, but can be as large as 50%.

IMPLICATIONS

Possible Canadian air quality objectives or standards for PM_{10} and $PM_{2.5}$ are currently under review, and in the United States a new $PM_{2.5}$ standard has recently been established. These actions are due, in part, to the increasing body of epidemiological evidence implicating airborne particles in human mortality and morbidity. Understanding the health effects of particles and devising monitoring programs and control strategies requires information on their composition and size distribution. The measurements collected as part of the Canadian National Air Pollution Surveillance (NAPS) program represent some of the most long-term, geographically diverse $PM_{2.5}$ data in the world and thus provide valuable insights into the behavior of $PM_{2.5}$. This paper focuses on temporal and spatial patterns in $PM_{2.5}$, including trends, diurnal variations, urban-rural differences, and within-city variations.

INTRODUCTION

Recent health research from many areas of the world indicates that airborne particulate matter (PM) is an important and pervasive air quality problem. Particles have been associated with a variety of health endpoints in a range of epidemiological studies.[1–3] Consequently, possible Canadian Air Quality Objectives for PM_{10} and $PM_{2.5}$ are under review,[4] and a new $PM_{2.5}$ standard has recently been established in the United States. Understanding the health effects of particles and devising control strategies requires spatial and temporal information on their mass concentration, composition, and size distribution.

Up to 14 years of every sixth-day 24-hr measurements of PM_{10}, $PM_{2.5}$, sulfate (SO_4^{2-}), and other trace elements are available for 19 Canadian locations.[5] A real-time PM monitoring network has also been established during the last five years.[6] Throughout Canada there are currently 34 tapered

element oscillating microbalances (TEOMs) measuring $PM_{2.5}$ and 43 TEOMs measuring PM_{10}. The purpose of this paper is to examine small- to large-scale spatial differences in PM in Canada and to discuss the temporal behavior of PM, including Canadian urban trends during the past 10 years. Due to the increase in the use of $PM_{2.5}$ TEOMs, this paper also includes an initial comparison of 24-hr gravimetric $PM_{2.5}$ measurements and TEOM measurements from the monitoring site with the largest amount of data.

DATA AVAILABILITY AND MEASUREMENT METHODS

The overall scope of Canadian PM monitoring activities is described in Dann and Brook[6] and in the Canadian PM Science Assessment document.[7] The manual measurements presented in this paper, which are part of the National Air Pollution Surveillance (NAPS) network, were collected on Teflon filters using Graseby-Anderson dichotomous samplers (dichot) as described in Brook et al.[5] After gravimetric analysis, chemical analyses of these filters were conducted using energy-dispersive X-ray fluorescence (EDXRF) and ion chromatography (IC).[5] The locations of the NAPS sites are shown in Figure 1. The majority of the sites are in urban areas; however, Egbert, Sutton, Walpole Island, and Kejimkujik are rural. There are two sites in Montreal, Toronto, Windsor, and Vancouver. All TEOM measurements were collected using inlets with either 10-µm or 2.5-µm cut-points (Rupprecht & Patashnick Co., Inc.). The instrument operating temperature was set to 50 °C. The TEOM measurements presented here were collected in Toronto at a provincially operated monitoring site (Evans or Toronto 2) that is located ~75 m south of a major highway. In order to examine particle composition and to evaluate the relationship between TEOMs and the existing NAPS manual method, a dichot was also located at Evans Avenue.

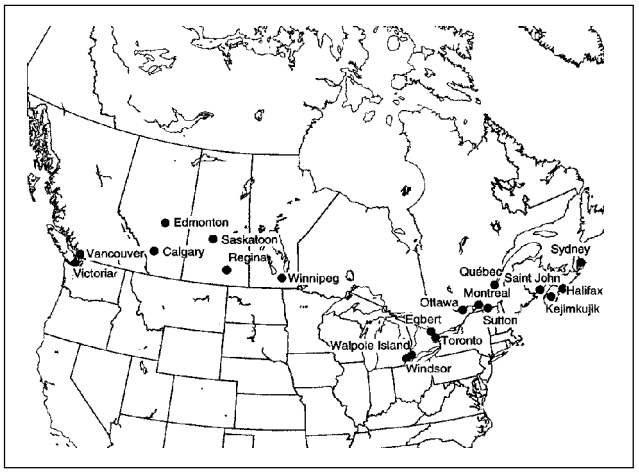


Figure 1. Location of NAPS network PM monitoring sites. Only data from selected sites are presented in this paper.

RESULTS

General Overview of the Network Measurements

During the 1990–1995 period, mean 24-hr $PM_{2.5}$ concentrations across Canada ranged from 5 to 20 µg m^{-3}, depending on site and season. A box plot showing summer and winter concentrations by site is presented in Figure 2. Levels were higher in the urban areas compared to the rural sites, with the highest levels in Hamilton (an industrial town) and Windsor (near Detroit industry). The location with the highest concentrations (not shown in Figure 2) was located at a second site in Montreal (Montreal 2, see Figure 4a), which is near a large amount of traffic.

In Figure 2 there are visible differences in the mean summer and winter $PM_{2.5}$ levels, but there is overlap in the mass distributions with a relatively similar amount of variability in both seasons. The sites with the largest differences in mean $PM_{2.5}$ between the summer and winter were Saint John, Kejimkujik, Toronto, Hamilton, Windsor, and Walpole Island. During the summer, mean $PM_{2.5}$ levels were higher over eastern Canada compared to the west. This is because in the summertime SO_4^{2-} increases dramatically at the eastern sites. East-to-west differences in mean $PM_{2.5}$ levels are less in the winter due to decreases in the eastern concentrations of SO_4^{2-} and to relatively large increases in $PM_{2.5}$ in Vancouver and Victoria. A number of factors could be leading to the wintertime increases in $PM_{2.5}$. Mixing heights are usually lower and energy consumption for heating is higher. Wood burning is also more prevalent, and a higher percentage of some semi-volatile species, such as nitrate (NO_3^-) and organic carbon, may be found in the particle phase.

On average across all sites, $PM_{2.5}$ explains 49% of the PM_{10}.[5] However, there is a considerable amount of variability in this percentage among sites and between individual

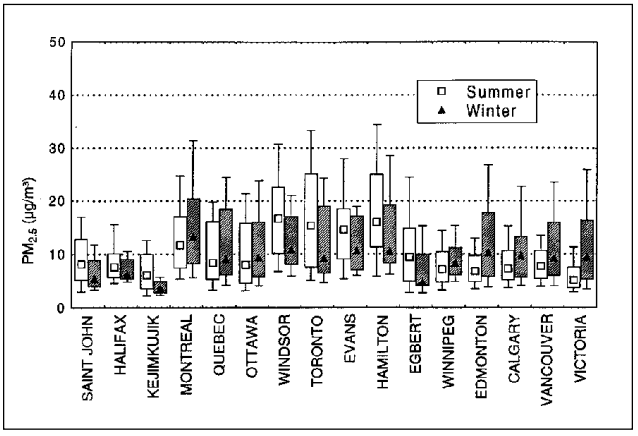


Figure 2. Winter (December, January, February) and summer (June, July, August) $PM_{2.5}$ concentrations (µg m^{-3}) for NAPS sites (all available 1-in-6 data from December 1989 to February 1998 are included). Plots show median 24-hr concentration (symbols) and 10th, 25th, 75th, and 90th percentile values.

observations. At the urban locations, the mean ratio of $PM_{2.5}$ to PM_{10} ranges from around 0.38 in the prairies to 0.59 on the west coast. The mean ratio ranges from 0.60–0.65 at the rural sites as well as at the Saint John site.

Figure 3, which is described in detail in Brook et al.,[5] shows that on average the major inorganic ions of SO_4^{2-} and NO_3^- (this component in the figure also includes ammonium [NH_4^+]) account for about 20–45% of the $PM_{2.5}$, depending upon the site. The majority of this fraction is from SO_4^{2-} and NH_4^+.[5] Although the contribution from NO_3^- is expected to be less than from SO_4^{2-}, NO_3^- measurements collected in Windsor over multiple seasons using an annular denuder system (ADS) showed that the dichot measurement technique as employed routinely in NAPS captures only about 10–15% of the particle NO_3^- determined using the ADS.[8] Thus, NAPS fine particle measurements significantly underrepresent NO_3^-. Correcting the SO_4^{2-}/NO_3^- component for this problem does not diminish the relative dominance of SO_4^{2-}, but it does increase the magnitude of the component by about 10% compared to what is shown in Figure 3.[8]

Figure 3 also shows that soil-based inorganic chemicals generally only account for 5–15% of the $PM_{2.5}$ and are most important to the overall mass in Winnipeg and least important in Toronto. Sodium chloride (NaCl) explains a larger portion of the $PM_{2.5}$ at the sites near salt water, but it still only accounts for a maximum of 5% of the mass. The remaining portion of the $PM_{2.5}$ is assumed to be associated with carbon species.[5] After adjusting Figure 3 for NO_3^- losses, as indicated above and in Brook and Dann,[8] it is estimated that the carbon fraction ranges from 55 to 30% of the total mass, depending upon the site. There is a greater percentage of carbon-related fine particles at the western sites (~62%) compared to those in the east (40–46%), particularly during the winter.

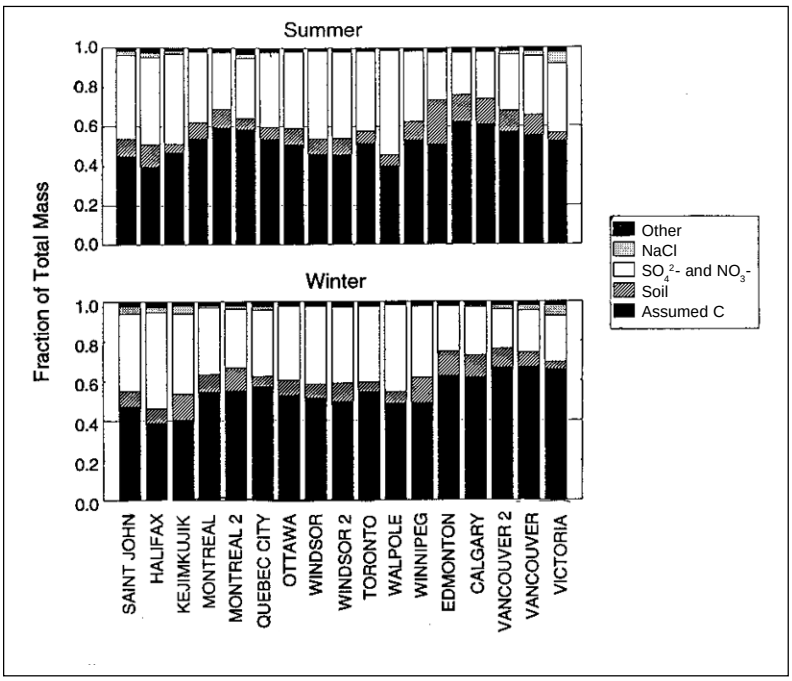
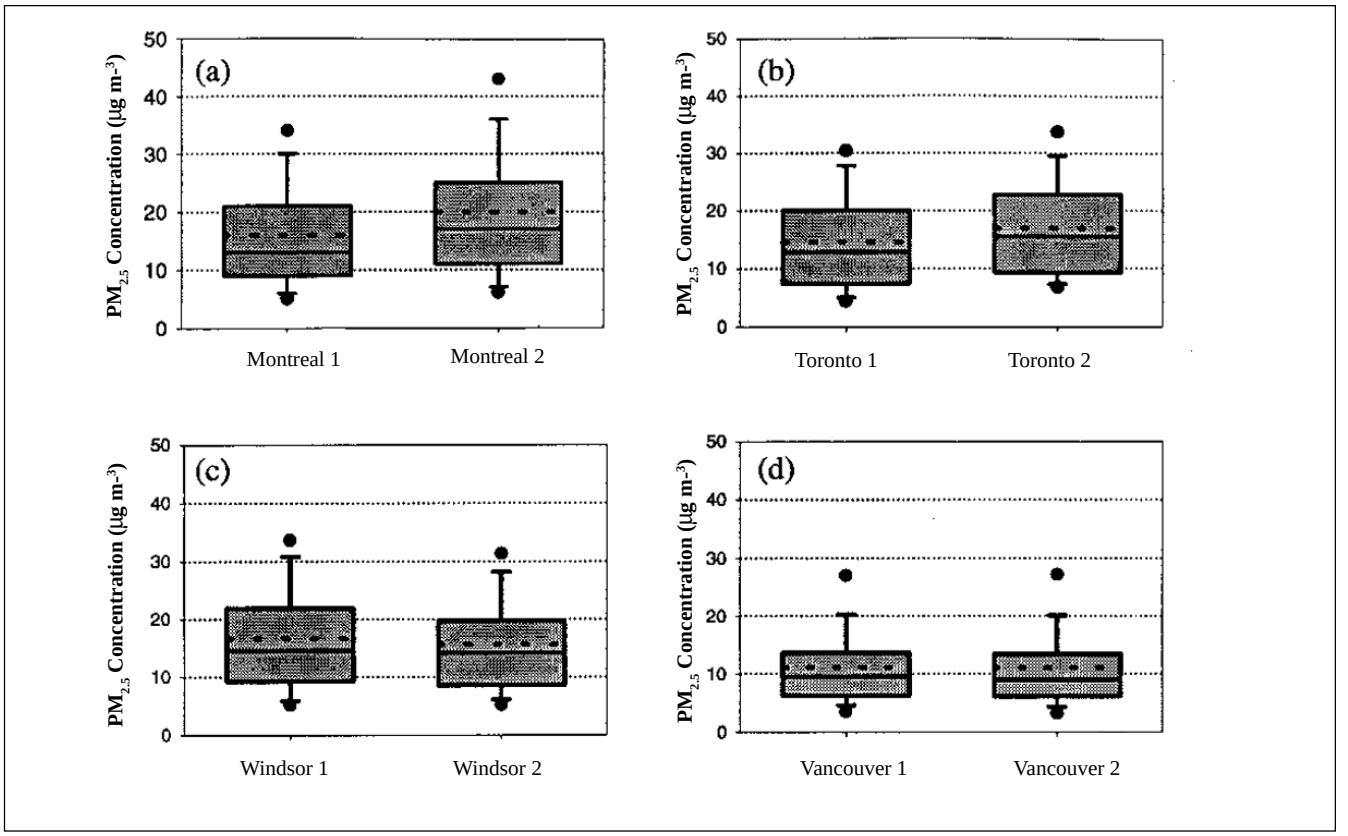


Figure 3. Relative proportions of the main chemical species making up $PM_{2.5}$ at the Canadian monitoring sites.

Spatial Variability and the Influence of Local Sources and Urban Areas

Broad geographical differences in $PM_{2.5}$ were highlighted above. There were also small-scale (i.e., within-city) differences in concentrations, depending upon measurement location and the local sources in the vicinity. Figures 4a–4d compare $PM_{2.5}$ measurements collected at pairs of sites within the same city. The greatest differences were between the pairs of Toronto and Montreal sites, where the mean concentrations differed by 2.3 μg m^{-3} and 4.0 μg m^{-3}, respectively. In both cities, vehicular traffic is believed to be the main source leading to the differences between sites since in each case one of the sites is near a major highway (i.e., Toronto 2 and Montreal 2). The differences between sites are statistically significant ($p < 0.01$) in each of the cities shown except for Vancouver. These within-city differences highlight the difficulty in determining where $PM_{2.5}$ sites or PM_{10} sites should be placed for compliance/regulatory monitoring purposes. Areas with higher levels cannot necessarily be excluded because there are likely to be people exposed to such levels. Use of multiple sites located in a range of settings appears to be the best approach for estimating exposure to ambient $PM_{2.5}$ and for obtaining a true measure of whether an area is in attainment or not. However, criteria need to be established to guide selection of these sites so that areas with higher levels are not systematically avoided. It is likely that even greater differences than those shown in Figures 4a–4d could be found within most cities if sampling locations were selected to maximize and minimize influence from local sources.

Not surprisingly, PM_{10} levels differ even more between sites within a city. Figure 5 shows the difference in PM_{10} between the two Montreal locations. The average concentration varies by 14 μg m^{-3}. Portable PM_{10} monitoring using MiniVol samplers (AirMetrics, Springfield, OR) was conducted in the vicinity of each of these two sites for 34 days in July and August 1996. For this relatively short-term pilot study, two contrasting areas in Montreal were examined, and the results are summarized in Figure 6. For the first 18 days, seven additional PM_{10} samplers were situated about 4 km northeast of the Montreal 1 site in an area that is predominantly residential. The sites were arranged in a circular pattern



Figures 4a–4d. Comparison of $PM_{2.5}$ concentrations between sites within the same city. Toronto 2 corresponds to the Evans site shown in Figure 2; and Montreal 1 is the same as Montreal in Figures 2 and 3. The distributions were determined from days when both sites were operating. Shown are the 95th, 90th, 75th, 50th, 25th, 10th, and 5th percentiles as well as the mean concentration (dashed line). The solid line represents the median value.

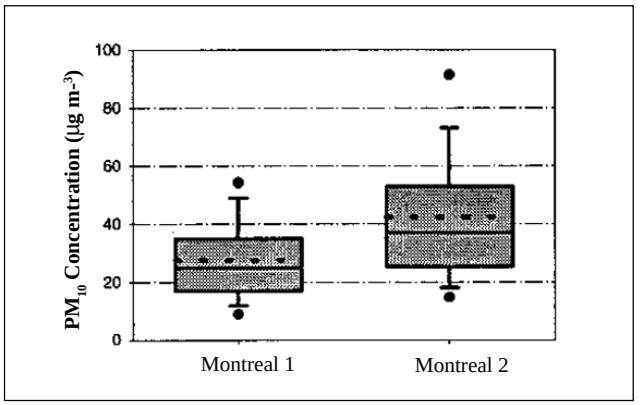


Figure 5. Comparison of the PM_{10} distributions between the two Montreal sites.

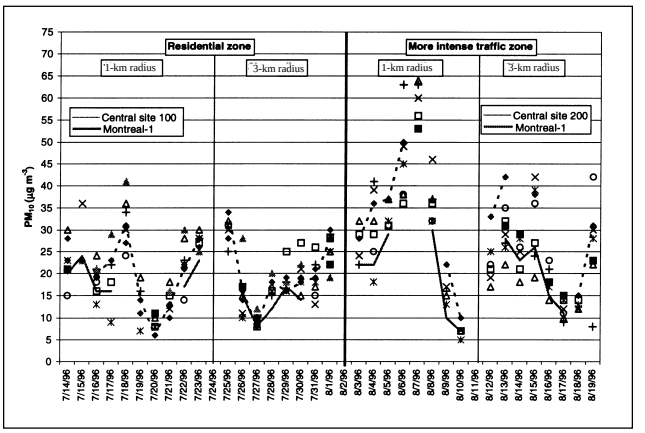


Figure 6. Spatial variations in PM_{10} across multiple sites in Montreal during July and August 1996. The solid line corresponds to the Montreal 1 site, and the dashed line corresponds to the Central site during the four different periods. Symbols represent the concentrations observed at temporary sites placed in different regions relative to the Central site and at either a 1- or 3-km radius.

around an established monitoring site (Central site 100—dashed line). For the first eight days the radius of this circle was 1 km, and for the second eight days the radius was increased to 3 km. Some of the samplers were on quiet streets and others were along busier streets, but not nearly as busy as the highway near the Montreal 2 site. The PM_{10} concentrations observed at the Montreal 1 site (solid line) and at each of the additional locations are shown in the first two sections of Figure 6. For the second 16 days (second two sections in Figure 6), five additional PM_{10} samplers were situated around the Montreal 2 site, which also corresponds to Central site 200 (dashed line). Within this area there is a major highway running north-south, light industry, and commercial and residential areas. Similar to the first 16 days, the samplers were arranged in a circle, first at a 1-km radius and then for the second eight days at 3 km around Montreal 2.

Figure 6 shows that, as suggested in Figure 5, there can be considerable spatial variability in PM_{10} concentrations, depending on the day of measurement. During the study, the largest daily difference between sites was about 20 μg m^{-3}. Comparison of the Montreal 1 measurements (solid line) to the range of other measurements (various symbols) and to the Montreal 2 measurements (dashed line for second half) shows that for the first 18 and last eight days, the Montreal 1 concentrations tended to be within the range of concentrations observed at the extra sites. In addition, for the last eight days the Montreal 2 concentrations tended to be greater than those at the extra sites and at Montreal 1. For the third portion of the study (August 3–10), the range of concentrations at the extra sites, which were all placed about 1 km from Montreal 2, encompassed what was observed at Montreal 2, but they were almost all greater than what was observed at Montreal 1. Although they are based upon a limited set of measurements, these results suggest that the high levels observed at Montreal 2 may extend to 1 km but generally do not extend to 3 km from the site. They also suggest that the levels at Montreal 1 are representative of more areas in Montreal. However, measurement across a wider range of areas, over a longer time period, and with the inclusion of $PM_{2.5}$ are necessary in order to assess, with more certainty, the overall representativeness of the Montreal 1 site and the zone of influence of local sources.

Urban influence on $PM_{2.5}$. Urban $PM_{2.5}$ concentrations reflect the local or urban emissions plus the incoming background air concentrations. The rural sites of Kejimkujik, Sutton, and Egbert are close enough to Saint John, Montreal, and Toronto, respectively, to estimate the local/urban influence on $PM_{2.5}$ by comparing urban and rural measurements. $PM_{2.5}$ and SO_4^{2-} concentrations for each of these pairs of sites are shown in Table 1. This comparison has been refined by using air-mass back-trajectories to identify specific sampling days when the urban area may have influenced the nearby rural measurement (i.e., when the wind was blowing from the urban area to the rural site). These days were excluded from the calculation of the averages in Table 1. Comparison of these values indicates that for SO_4^{2-}, locally derived emissions increased the concentrations from a factor of 1.28 to greater than a factor of two, depending upon location. Similarly, 30 to 82% of the $PM_{2.5}$ is estimated to be generated locally. On average, the increase in $PM_{2.5}$ tends to be more than a factor of two larger than the increase in SO_4^{2-}. For example, in Toronto 1.3 μg m^{-3} of SO_4^{2-} is estimated to be of local origin, while there is 4.0 μg m^{-3} more $PM_{2.5}$ as compared to Egbert. Assuming that the SO_4^{2-} was in the form of ammonium sulfate (NH_4SO_4), there is 2.2 μg m^{-3} of non-sulfur-based $PM_{2.5}$ generated locally. Much of this extra mass is believed to be carbonaceous material. Measurements are currently being collected to test this hypothesis.

Temporal Variability and Trends

Diurnal variations in $PM_{2.5}$ in Southern Ontario. Seasonal variations in $PM_{2.5}$ were discussed above. An analysis of differences in weekday versus weekend $PM_{2.5}$ concentrations across Canada can be found in Dann and Brook.[6] Thus, the focus of the discussion here is on TEOM measurements of diurnal patterns in $PM_{2.5}$. Possible limitations mainly due to the loss of semi-volatile material, which can be a problem with the use of the TEOM, are discussed next. The two locations with the longest TEOM record for $PM_{2.5}$ are Evans (Toronto 2) and Egbert. A TEOM with a 2.5-μm-cyclone inlet has been in operation at Evans since July 1995 and at Egbert since March 1996. Average $PM_{2.5}$ concentrations by hour are shown for each site for the warm season (May–September) and the cold season (October–April) in Figures 7a and 7b. There was a clear diurnal pattern in the concentrations at Evans, with a peak in the morning around 0800 EST; concentrations were lowest after midnight, at around 0200–0400 EST. A similar diurnal pattern was not evident at Egbert—in contrast, the levels were slightly higher in the afternoon, especially in the warm season.

Diurnal variations in emissions and meteorological and chemical processes are each partly responsible for the behavior shown in Figures 7a and 7b. The sharp rise in $PM_{2.5}$ in the morning at Evans is a function of an increase in emissions (i.e., rush-hour traffic) and closely resembles the increase observed in CO concentrations shown in both figures. Most of these particles are likely to be of primary origin (i.e., assumed to be emitted directly as particles).

Table 1. $PM_{2.5}$ and SO_4^{2-} concentrations (μg m^{-3}) at nearby urban-rural pairs of sites and estimates of the local contributions. Percent contribution [100x(urban-rural)/rural)] from local/urban sources shown in parentheses.

Site Pair	Rural Concentrations		Urban Concentrations		Estimated Local Input	
	SO_4^{2-}	$PM_{2.5}$	SO_4^{2-}	$PM_{2.5}$	SO_4^{2-}	$PM_{2.5}$
Saint John/Kejimkujik	1.4	4.7	2.9	8.5	1.5 (107)	3.9 (82)
Montreal/Sutton	2.9	12.6	3.7	16.3	0.8 (28)	3.7 (29)
Toronto/Egbert	3.7	10.0	5.0	14.0	1.3 (35)	4.0 (40)

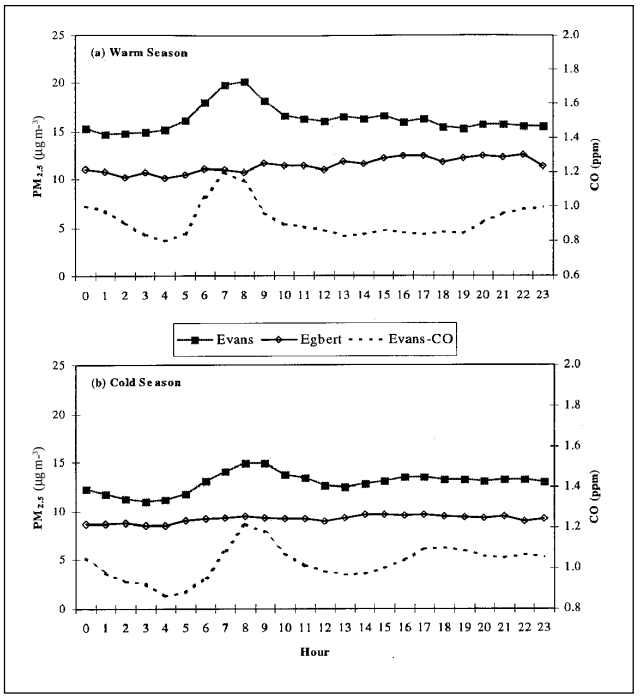


Figure 7. Average diurnal variation in $PM_{2.5}$ and CO at Evans (Toronto 2) and $PM_{2.5}$ at Egbert.

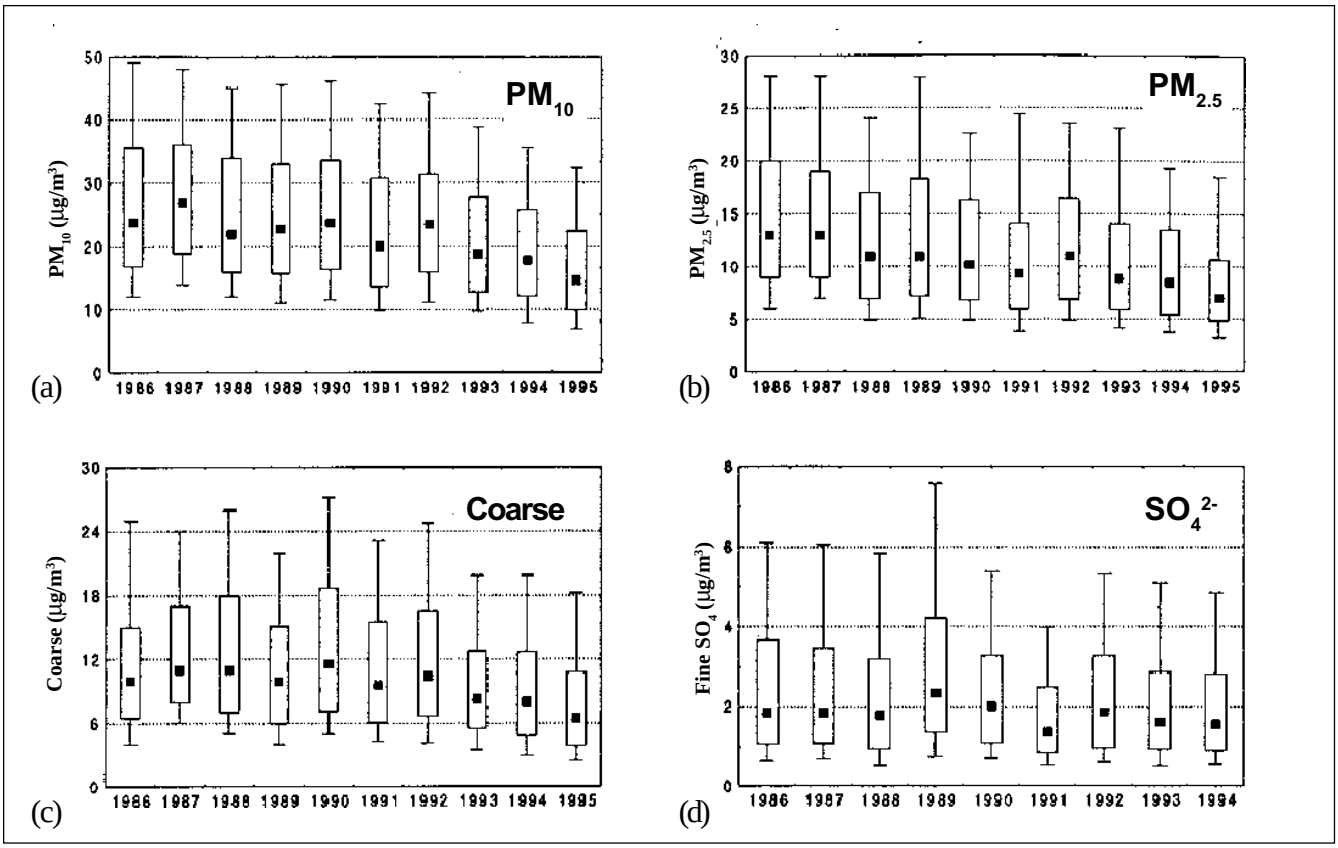
The decrease in the morning peak occurs due to the end of rush hour, but is also due to vertical mixing as the nocturnal boundary layer breaks up. We hypothesize that mixing is at least as important as the drop in emissions because of the abruptness of the change in concentration and because of the one-hour shift in the time between the warm and cold season (i.e., weaker solar heating in the cold season results in a later breakup of the low-level inversion). However, there is also a change to daylight savings time (DST) in the warm season which causes a shift in the timing of rush hour relative to EST. The lack of a distinct peak during the evening/afternoon rush hour compared to the morning also suggests that the mixing height plays an important role. Further analyses are needed to better understand the processes leading to the diurnal pattern in $PM_{2.5}$.

$PM_{2.5}$ variations in the regional background air, as evidenced by the observations at Egbert, appear to be more influenced by photochemical production. The warm season peak in the afternoon is likely a result of increased concentrations of secondary sulfate and secondary organic aerosols. In addition, transport of the higher particle levels produced in the morning in the Toronto-Hamilton areas could also play a role, both as primary and secondary particles, in the elevated afternoon concentrations at Egbert.

The morning increase in $PM_{2.5}$ at Evans also provides some insight into the amount of urban fine particles generated by local emissions. In the morning, vehicles and other sources are emitting particles and their precursors into a relatively shallow atmospheric layer. The increase in the TEOM concentrations in the morning is predominantly due to these emissions, assuming that the height of the boundary layer does not change dramatically between 0400 and 0800 hr EST. In the warm season, these local emissions lead to about a 4.9-μg m^{-3}, or a 32%, increase in $PM_{2.5}$. In the cold season this increase is 34%. These estimates of local influences are slightly smaller than the results shown in Table 1 discussed above. However, given the uncertainties involved in these analyses (e.g., variations in mixing height and site representativeness), these two independent estimates of local influences are reasonably, if not surprisingly, consistent.

10-Year trends in $PM_{2.5}$ and PM_{10}. There are 10 urban sites in the NAPS network with available complete PM_{10} and $PM_{2.5}$ data for a 10-year (1986–1995) time period. Composite results (data for all 10 sites combined) for PM_{10}, $PM_{2.5}$, coarse mass (2.5–10 μm), and fine SO_4^{2-} are provided in Figures 8a–8d. From 1986 to 1995, composite median PM_{10} concentrations decreased from 24 to 14.5 μg m^{-3}. This decrease resulted from a 6-μg m^{-3} decrease in fine fraction mass and a 3.5-μg m^{-3} decrease in coarse fraction mass. At each of the 10 sites, there was a significant decrease in PM_{10} and $PM_{2.5}$ between 1986 and 1995 ($p < 0.01$; Mann-Whitney median test). Median fine sulfate concentrations declined only slightly (0.5 μg m^{-3}) in that time period, indicating that decreases in fine fraction mass were primarily due to decreases in other chemical constituents.

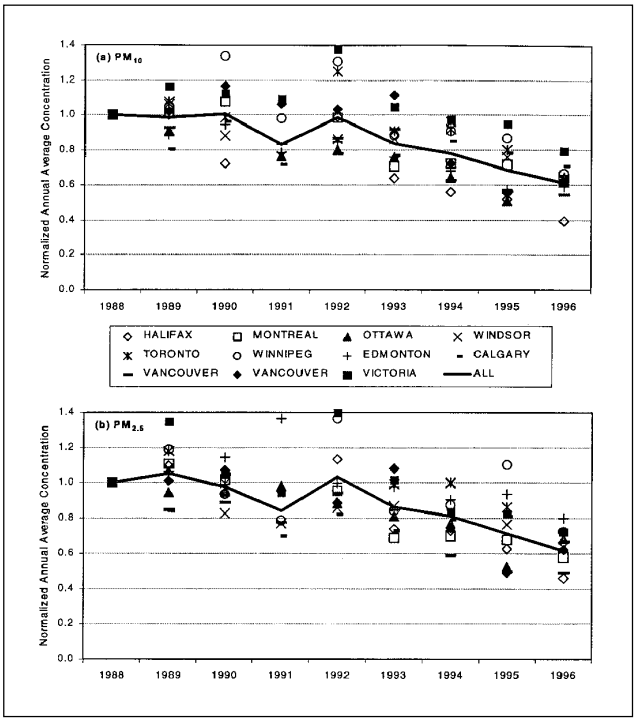
Widespread decreases in PM_{10} have also been observed in the United States. Darlington et al.[9] report that from 1988 through 1995, the average U.S. PM_{10} concentration decreased from 23 to 33%, depending upon corrections for background concentrations (i.e., the ratio of the 1995 and 1988 national average concentrations ranged from 0.77 to 0.67). Expressing the trends from the 10 Canadian sites in the same way shows that the composite decrease in PM_{10} was 32% (i.e., the PM_{10} ratio of 1995 to 1988 is 0.68). This is 9% greater than the change determined for the United States using a zero background estimation, but is within the range of values obtained when background levels are considered.[9] As shown in Figures 8a–8c, the PM_{10} changes at the Canadian sites were a result of both fine and coarse particle reductions. On a mass basis the fine particle levels decreased more, but on a percent basis a greater reduction occurred in coarse particles. Relative to 1988, the composite average $PM_{2.5}$ was 28% lower in 1995 and the coarse particle levels were 35% lower. Assuming that these trends are associated with changes in emissions would indicate that although source reductions in combustion-related particles and secondary particles occurred, there was a greater relative decline in emissions from sources associated with coarse particles, such as fugitive dust, road dust, and open burning.



Figures 8a–8d. Interannual variability and trends in PM_{10}, $PM_{2.5}$, coarse mass, and fine SO_4^{2-} concentrations (μg m^{-3}) for composite data from 10 urban sites. Plots show median 24-hr concentration and 10th, 25th, 75th, and 90th percentile values.

Year-to-year changes in average concentrations from 1988 to 1996, expressed relative to 1988 (i.e., as done by Darlington et al.[9]), are shown by site in Figures 9a and 9b. There was considerable variability among sites/cities in the interannual variability in both PM_{10} and $PM_{2.5}$. On average, across the country there was little change for the first five years, with sites experiencing decreases, increases, or both, depending upon the year. Much of this likely reflects meteorological variability. In contrast, for 1994–1996 the annual average PM_{10} was less than the 1988 average at all sites, and $PM_{2.5}$ behaved similarly at a large majority of the sites. The net result of this behavior is an overall decrease in the levels across Canada's major cities. However, individual sites in Figures 9a and 9b do deviate from this pattern. For example, annual average PM_{10} and $PM_{2.5}$ at the Calgary site decreased by about 20% in the first year and then remained relatively steady, bouncing above and below the 0.8 line. Meteorological variations could have caused these year-to-year changes, including a relatively high year such as 1988.

To smooth out the influence of meteorological variability, the fine and coarse mass data were averaged across four-year periods. These results are shown in Table 2. To maximize the amount of data included, the initial averaging period was 1987–1990 and the final averaging period



Figures 9a and 9b. Interannual variability in PM_{10} and $PM_{2.5}$ concentrations at selected Canadian sites between 1988 and 1996. Concentrations are normalized to 1988 values.

Table 2. Percent changes in fine mass and coarse mass at the Canadian urban monitoring locations.

City	Fine			Coarse		
	87–90	93–95	% Change	87–90	93–96	% Change
Halifax	13.4	8.5	-37%	11.3	5.7	-49%
Montreal	18.8	11.5	-39%	13.3	8.9	-33%
Ottawa	13.5	9.3	-31%	12.9	7.9	-39%
Windsor	19.8	16.1	-19%	15.5	11.9	-23%
Toronto	15.7	14.0	-11%	12.1	9.8	-19%
Winnipeg	10.0	8.0	-20%	19.3	13.0	-33%
Edmonton	10.6	8.3	-22%	17.9	10.7	-40%
Calgary	11.3	8.6	-24%	16.2	13.8	-15%
Vancouver	14.0	9.2	-35%	9.9	7.5	-24%
Vancouver	14.5	9.7	-33%	9.2	6.1	-34%
Victoria	11.4	8.0	-30%	5.9	6.0	1%

was 1993–1996. There were relatively large decreases in both the fine and coarse mass at all the sites except coarse mass at Victoria, which did not change. For fine mass, the decreases were smallest in southern Ontario and largest in Montreal. The decreases were also relatively large in Vancouver and Victoria. There was a large decrease observed in Halifax as well, but due to a local source the dichot was moved to another location in 1994, which probably influenced the trend. With the exception of Victoria and Halifax, there is similar site-to-site variability in the coarse mass trends. The lack of a trend at Victoria was influenced by the greater importance of sea salt,[5] which is likely to exhibit less change from year to year. The large decrease in Halifax was probably due to the relocation of the dichot in 1994.

Because of the lack of adequate emission inventory data for PM, it is currently difficult to identify the causes of the decreases in PM_{10} and $PM_{2.5}$ concentration at the Canadian urban sites. However, transportation sources are known to be major contributors to PM loading at urban sites[10]; thus, improvements in emission control technologies for light- and heavy-duty vehicles and the introduction of cleaner fuels (elimination of leaded gasoline and reduced sulfur in diesel) have likely played a role in the observed trends.

Comparison of the TEOM and Dichot $PM_{2.5}$ Measurements

The amount of monitoring and research on $PM_{2.5}$ and its chemical constituents is increasing in Canada. Consequently, several $PM_{2.5}$ TEOMs have been installed and the inlets on a number of the pre-existing PM_{10} TEOMs have been changed. The manual dichot network also continues to be operated, and thus data on particle composition are available. There are a number of advantages to using TEOMs:

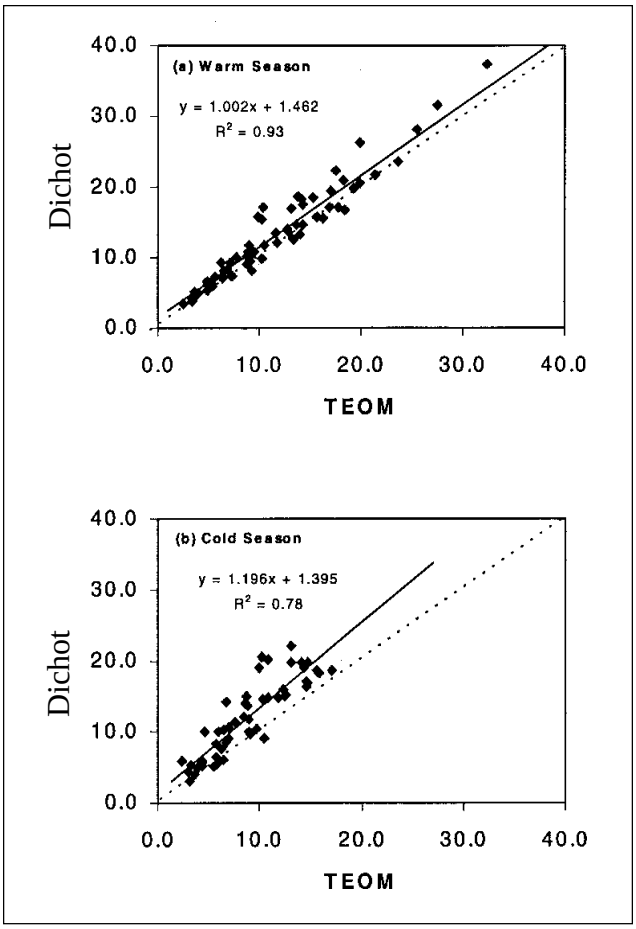
- Daily mass measurements are available for epidemiological research.
- Diurnal patterns and hourly maximum values can be studied.
- $PM_{2.5}$ concentrations are available in nearly real time to support urban smog advisories/forecasts and rapid determination of the air quality index.

However, as with all methods for measuring the mass of the aerosol suspended in the air, there are uncertainties in the TEOM measurements.[11] Some of these uncertainties are associated with the loss of semi-volatile chemical constituents arising from the need to maintain a constant temperature for the microbalance, which is typically set to 50 °C.

At a number of locations, TEOMs are collocated with dichots and ongoing comparisons of these two different $PM_{2.5}$ measurements are being conducted. This provides a link between past measurements and future data and offers insight into the appropriateness of studying spatial patterns in $PM_{2.5}$ using measurements from dichots and TEOMs. In addition, the TEOM mass measurements can be evaluated relative to sampling and gravimetric approaches that are used in Canada.

The largest amount of collocated dichot and TEOM data are from Evans, where both instruments have been operating since July 1995. Due to the sampling schedule of the dichot, comparison of 24-hr values is possible every sixth day. Figures 10a and 10b show how the two measurement methods compare in the warm and cold seasons. The TEOM concentrations in these plots have not been adjusted by the manufacturer-recommended adjustment factors necessary for the TEOM to qualify as an equivalent PM_{10} method.[12] This allows for a direct comparison between the two measurement techniques, as was done by Allen et al.[11] Furthermore, PM_{10} adjustments may not be appropriate for $PM_{2.5}$.

With the exception of a relatively small number of measurements, the TEOM $PM_{2.5}$ values are lower than the dichot values during both the warm and cold seasons. Thus, as expected, there is systematic loss of mass from



Figures 10a and 10b. Warm and cold season comparison of 24-hr TEOM and dichot $PM_{2.5}$ concentrations (μg m^{-3}) collected at Evans Avenue (Toronto 2) in 1995 and 1996 and during two 2-week intercomparison studies in 1998.

the TEOM measurements. There is considerably better agreement in the warm season compared to the cold season. The regression lines and the R^2 values are shown on Figures 10a and 10b.

During the warmer months, the apparent loss of $PM_{2.5}$ material using the TEOM compared to the dichot is relatively small, about 12% (mean TEOM-to-dichot ratio of 0.88). There is also some evidence on Figure 10a that there is less loss when the $PM_{2.5}$ concentrations are smaller. This tendency was also evident in the cold season (Figure 10b). In contrast to the warm season, however, the magnitude of the losses at higher concentrations was larger during the cold season. The average TEOM-to-dichot ratio in the cold season was 0.77, but in the worst case the TEOM 24-hr concentration was only 50% of the dichot concentration.

The greater overall loss in the cold season is due to the existence of greater amounts of particle nitrate during the colder months.[8] A greater amount of particle phase semi-volatile carbon probably had an influence as well. The slope of the regression line in Figure 10b indicates that in the colder months the mass contribution by semi-volatile species increases as $PM_{2.5}$ increases. The losses due to this behavior are smaller using the manual approach because the dichot collects the particles at ambient temperatures, and between the time of collection and gravimetric analysis the filters are generally kept sealed and at ≤ 25 °C. In an attempt to reduce the losses associated with TEOM measurements, the instrument operating temperatures throughout the national network were reduced to 40 °C in late 1997 and early 1998. The impact of this change will be assessed in the future.

CONCLUSIONS

Up to 14 years of Canadian 24-hr manual particle measurements and two years of hourly measurements have been examined in this paper. The main results and conclusions are summarized below.

A large majority of the 24-hr $PM_{2.5}$ concentrations in Canada are below the U.S. 24-hr standard. For certain years and sites, the annual average $PM_{2.5}$ as measured by the dichot sampler is above the U.S. annual standard of 15 μg m^{-3}.

Multiple monitoring sites within cities show that there can be significant differences in $PM_{2.5}$ among measurement locations. The use of multiple sites located in a range of settings appears to be the best approach for estimating exposure to ambient $PM_{2.5}$ and for obtaining a true measure of whether $PM_{2.5}$ and/or PM_{10} levels in a given area exceed current standards or objectives.

Even greater differences have been observed between the $PM_{2.5}$ concentrations within a city and $PM_{2.5}$ in nearby rural areas. This indicates that 30–80% of the $PM_{2.5}$ measured in a city is due to local/urban sources. Diurnal patterns in $PM_{2.5}$, which show a distinct rush-hour peak in concentrations, also suggest that a significant fraction of the urban $PM_{2.5}$ is local in origin. $PM_{2.5}$ and PM_{10} levels at Canadian urban sites decreased from the mid-1980s to the mid-1990s. From 1988 to 1995, the composite annual average PM_{10} concentration decreased by 32%, and $PM_{2.5}$ decreased by 28%.

Comparison of the largest collocated TEOM and dichot $PM_{2.5}$ data set shows reasonable agreement in 24-hr $PM_{2.5}$ concentrations, especially during the warmer months. However, there is evidence that the TEOM indicates lower concentrations than the manual dichot method due to the loss of $PM_{2.5}$ material. This loss is greater in the cold season than in the warm season.

In order to ensure compatibility with U.S. observations, it will be important to compare the measurement methods being employed at the Canadian sites with the current U.S. Environmental Protection Agency Federal Reference Method for $PM_{2.5}$. This work is currently being carried out at the Egbert and Evans Avenue sites.

With the possibility of future Canadian Air Quality Standards for PM_{10} and $PM_{2.5}$, increasing attention will be

placed on emission reduction strategies for Canadian sources and possible international solutions to the $PM_{2.5}$ problem. The existing monitoring program and periodic short-term research studies will continue to play an important role in health effects research, in identifying effective abatement strategies, and in the refinement of emission inventories. The recent expansion of the continuous PM monitoring networks will also allow greatly improved reporting of daily air quality results in urban areas.

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